

Testing similarity of competing risks models by comparing transition probabilities

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Abstract

Assessing whether two patient populations exhibit comparable event dynamics is essential for evaluating treatment equivalence, pooling data across cohorts, or comparing clinical pathways across hospitals or strategies. We introduce a statistical framework for formally testing the similarity of competing risks models based on transition probabilities, which represent the cumulative incidence of each event over time. Our method relies on a parametric modeling approach, motivated by the need to develop reliable inference procedures for small sample settings in which nonparametric estimators become unstable or infeasible. We consider a maximum-type distance between the transition probability matrices of two competing risks processes and develop a novel constrained parametric bootstrap test to evaluate similarity under both administrative and random right censoring. We theoretically establish the asymptotic validity and consistency of the bootstrap test. Through extensive simulation studies, we show that our method reliably controls the type I error and achieves higher statistical power than existing intensity-based approaches. Applying the framework to routine clinical data of prostate cancer patients treated with radical prostatectomy, we identify the smallest similarity threshold at which patients with and without prior in-house fusion biopsy exhibit comparable readmission dynamics. The proposed method provides a robust and interpretable tool for quantifying similarity in event history models.

Keywords: Similarity testing, Parametric bootstrap, Competing risks models, Transition probabilities, Clinical pathways, Routine clinical data

1 Introduction

Understanding how patients progress through different stages of treatment is a key objective of medical research and healthcare service analysis. With the increasing availability of routinely collected hospital data, it has become possible to reconstruct treatment pathways that describe the sequence of events a patient experiences during treatment, such as surgeries, complications, or readmissions. These treatment pathways capture the dynamics of clinical care processes and often differ between patient groups, hospitals, or treatment strategies. Comparing such pathways provides valuable insights into whether different medical practices lead to similar outcomes or whether a strategy changes the patient's further course. An example question would be whether long-term event patterns, such as readmission risks or complications, depend on specific diagnostic procedures or whether they are comparable. The possible identification of such similarities is not only important for clinical interpretation, but also determines whether data from different patient cohorts can be combined for further statistical analysis (Binder et al., 2022).

To address these questions, patient pathways can be modeled as event history processes that evolve over time. A natural framework for this is provided by competing risks models (or more general multi-state models), which describe how individuals transition between discrete health states until an absorbing outcome occurs (Andersen et al., 2012; Andersen and Keiding, 2002; Beyersmann et al., 2012; Putter et al., 2007). Each possible event type, such as readmission for a specific cause, represents a competing risk, and the model quantifies the rate or probability of transitions between states. Within this setting, assessing whether two groups of patients, such as those who received a particular diagnostic intervention and those who did not, share similar transition behavior becomes a problem of comparing two stochastic processes. However, formal statistical methods for testing *similarity* between such processes remain scarce, even though these comparisons are fundamental for evaluating equivalence between treatment strategies, care pathways, or healthcare systems. While competing risks analyses do offer a number of established hypothesis tests designed to detect significant *differences* between groups (Lin, 1997; Williamson et al., 2007; Bakoyannis, 2020; Lyu et al., 2020; Sestelo et al., 2024), these methods are not aimed at evaluating equivalence. On the other hand, similarity testing has found considerable interest in other applications such as equivalence testing of pharmacokinetic and pharmacodynamic models (Gsteiger et al., 2011; Kaneko, 2025), the comparison of dose response profiles (Liu et al., 2009; Möllenhoff et al., 2021; Dette et al., 2025), the assessment of consistency in clinical trials (Grill et al., 2020) or the validation of similarity between quantile regression models in ecological applications (Cade, 2011). Most of these approaches were inspired by equivalence testing concepts from pharmacokinetics, where bioequivalence between two drug concentration profiles is determined based on minimal differences (Hauschke et al., 2007).

In recent years, methodological developments have introduced formal approaches for testing the similarity between competing risk models. Binder et al. (2022) proposed a framework for comparing competing risk models based on cause-specific transition intensities. In this framework, each patient group is treated as a Markov process with constant hazard and continuous time. Because these authors, and we in this paper as well, were particularly interested in data analysis for small sample sizes, where nonparametric approaches typically are less efficient (see e.g. Section 3.1 in Jullum and Hjort, 2019), they made parametric assumptions about the intensities to address the small sample issue. Specifically, they defined a similarity test that evaluates whether the transition intensities of two groups differ by more than a predefined threshold. Their testing procedure is not based on the intersection-union principle as the classic two one-sided test (TOST) in bio-equivalence studies, but involves a novel bootstrap approach, which mimics the (asymptotically) uniformly powerful test (see Romano, 2005). This superiority can also be observed for finite samples (see also Möllenhoff et al., 2022, for a similar observation in bio-equivalence problems). Möllenhoff et al. (2024) extended this approach to parametric intensity models beyond the case of constant hazard values, including Gompertz and Weibull distributions to better represent realistic event time patterns. Additionally, they standardized comparisons across different causes of error using a global supremum-type test statistic. This improves the informative power and interpretability compared to separate tests for each cause. These studies established the first systematic methods for testing similarity in event history models and demonstrated that intensity-based bootstrap tests can effectively control type I error and achieve high power in many scenarios.

Despite these advances, intensity-based similarity measures have inherent limitations. Transition intensities describe instantaneous risks and thus capture the local behavior of the process, but they do not directly reflect the overall probabilities with which each event will occur within a given time horizon, the latter which are often more relevant and intuitive in a medical context. The probability that a particular outcome will occur by a given time is often more interpretable than its instantaneous risk. Furthermore, two models may have similar intensities at most points in time but differ significantly in their cumulative risks, especially if the hazards vary over time or if the censoring patterns differ between groups. Consequently, a comparison based on tran-

sition probabilities, which summarizes the cumulative probability of each outcome, provides a more interpretable and potentially more sensitive measure of similarity between competing risk models (Andersen and Keiding, 2002; Geskus, 2015).

In this paper, we introduce a novel similarity testing framework for competing risks models that focuses on transition probabilities rather than transition intensities. By comparing the transition probability matrices of two competing risks processes over time, our method captures the full range of possible state transitions, enabling a comprehensive and interpretable assessment of model similarity. To quantify similarity between models, we define a metric that measures the maximum absolute deviation in transition probabilities across all causes and time points. Based on this metric, we construct a statistical test for the null hypothesis of model dissimilarity. To approximate the null distribution of the test statistic, we develop a constrained parametric bootstrap procedure designed to also accommodate right-censoring. The paper is structured as follows. In Section 2, we define the modeling setting and outline the algorithmic procedure for testing the global hypotheses for two different right-censoring schemes. In Section 3, we demonstrate the validity of the new approach and compare its performance to a previously proposed method (Möllenhoff et al., 2024). In Section 4, we explain the application example that originally motivated this research. Finally, we conclude with a general discussion.

2 Modeling setting and similarity test approach

2.1 Competing Risk Models and Estimation of Transition Probabilities

For modeling the treatment pathways observed for two different groups of patients we follow Andersen et al. (2012) and consider two independent Markov processes in continuous time

$$X^{(1)} = (X^{(1)}(t))_{t \geq 0}, \quad X^{(2)} = (X^{(2)}(t))_{t \geq 0} \quad (1)$$

with finite state space $\{0, 1, \dots, k\}$ (for a fixed $k \in \mathbb{N}$). State 0 corresponds to the starting state for every patient, that is, for $\ell = 1, 2$

$$\mathbb{P}(X^{(\ell)}(0) = 0) = 1,$$

while we identify the competing risks with the other states $1, \dots, k$. The probability for a transition of a patient from state 0 at time s to one of the other states $j \in \{1, 2, \dots, k\}$ at time $t > s \geq 0$ is given by

$$P_{0j}^{(\ell)}(s, t) = \mathbb{P}(X^{(\ell)}(t) = j | X^{(\ell)}(s) = 0), \quad (2)$$

and the cause-specific transition intensities from state 0 to state j of the process $X^{(\ell)}$ are defined by

$$\alpha_{0j}^{(\ell)}(t) = \lim_{\Delta t \rightarrow 0} \frac{P_{0j}^{(\ell)}(t, t + \Delta t)}{\Delta t}. \quad (3)$$

Both, the transition intensities

$$\alpha^{(\ell)} = \{\alpha^{(\ell)}(t) | t \geq 0\} = \{(\alpha_{0j}^{(\ell)}(t))_{j=1, \dots, k} | t \geq 0\}$$

as well as the transition probabilities

$$P^{(\ell)} = \{P^{(\ell)}(s, t) | t > s \geq 0\} = \{(P_{0j}^{(\ell)}(s, t))_{j=1, \dots, k} | t > s \geq 0\}$$

characterize the competing risks model $X^{(\ell)}$ and can therefore be used for measuring the similarity between $X^{(1)}$ and $X^{(2)}$. In contrast to Binder et al. (2022) and Möllenhoff et al. (2024), who discussed the similarity between competing risk models using measures based on transition intensities, we focus in this article on similarity measures based on transition probabilities. The

consideration of these quantities instead of intensities is motivated by the observation that transition probabilities provide a more natural description of the probabilities for phase transition than transition intensities. Therefore, test statistics based on transition probabilities might be able to capture the underlying truth of similarity of competing risks models better than test statistics based on transition intensities can. On the other hand, investigating the theoretical properties of similarity measures based on transition probabilities poses substantial methodological challenges as will be explained in the following discussion.

For a comparison with the work of Möllenhoff et al. (2024) and to keep the technical arguments to a minimum, we consider constant transition intensities throughout this paper. Additionally, we investigate the problem for similar censoring mechanisms as in Möllenhoff et al. (2024). That is, we investigate type I censoring at a fixed time $T \in \mathbb{R}_{>0}$ in this section as well as random right censoring in Section 2.2.

Under the assumption that all transition intensities are constant, the Markov processes are stationary and the transition probabilities in (2) depend only on the time difference $t - s$. Therefore, taking $s = 0$, we obtain

$$P_{0j}^{(\ell)}(t) = \mathbb{P}(X^{(\ell)}(t) = j | X^{(\ell)}(0) = 0) = -\frac{(-1 + e^{-\alpha_0^{(\ell)}t})\alpha_{0j}^{(\ell)}}{\alpha_0^{(\ell)}} \quad (4)$$

for $j = 1, \dots, k$, where

$$\alpha_0^{(\ell)} = \sum_{j=1}^k \alpha_{0j}^{(\ell)} \quad (5)$$

is the all-cause hazard and the second equality in (4) follows from Albert (1962). We will estimate the transition intensities by the maximum likelihood method. For type I censoring at time point $T \in \mathbb{R} > 0$, we consider the two competing risks models $\{X^{(1)}|t \in [0, T]\}$, $\{X^{(2)}|t \in [0, T]\}$ and note that a sample can be specified by the knowledge of the survival time of the individual and the target state of the transition. By Andersen and Keiding (2002), the distribution of the survival time $T^{(\ell)}$ of individuals in group $\ell = 1, 2$ is given by

$$S^{(\ell)}(t) := \mathbb{P}(T^{(\ell)} \geq t) = \exp\left(-\sum_{j=1}^k \alpha_{0j}^{(\ell)}t\right).$$

For sample sizes $n_1, n_2 \in \mathbb{N}$, let $X_1^{(1)}, \dots, X_{n_1}^{(1)}$ and $X_1^{(2)}, \dots, X_{n_2}^{(2)}$ be iid samples of the processes $X^{(1)}$ and $X^{(2)}$ in (1), respectively. They represent the healthcare pathways of two groups of patients. For each sample $X_i^{(\ell)}$, we observe the survival time and target state denoted by $(T_i^{(\ell)}, X_i^{(\ell)}(T_i^{(\ell)}))$. Then, as in Andersen et al. (2012), the log-likelihood function, based on the observations of group ℓ with type I censoring, is given by

$$\log \mathcal{L}^{(\ell)}(\alpha^{(\ell)}) = \sum_{i=1}^{n_\ell} \log(S^{(\ell)}(T_i^{(\ell)})) + \sum_{i=1}^{n_\ell} \sum_{j=1}^k I\{X_i^{(\ell)}(T_i^{(\ell)}) = j\} \log(\alpha_{0j}^{(\ell)}). \quad (6)$$

Maximizing the log-likelihood function yields the maximum likelihood estimators (MLEs)

$$\hat{\alpha}_{0j}^{(\ell)} = \frac{\sum_{i=1}^{n_\ell} I\{X_i^{(\ell)}(T_i^{(\ell)}) = j\}}{\sum_{i=1}^{n_\ell} T_i^{(\ell)}} \quad (7)$$

for the transition intensities $\alpha_{0j}^{(\ell)}$ with $j = 1, \dots, k$, $\ell = 1, 2$. In other words, we estimate the transition intensities by dividing the number of individuals in group ℓ that transitioned from state 0 to state j by the total survival time of all individuals in group ℓ .

An estimator of the transition probabilities can be constructed as a plug-in estimator exploiting equation (4). For that purpose, we define an estimator of the all-cause hazard in (5) by

$$\hat{\alpha}_0^{(\ell)} = \sum_{j=1}^k \hat{\alpha}_{0j}^{(\ell)}. \quad (8)$$

Then, the estimators of the transition probability functions are given by $\{\hat{P}_{0j}^{(\ell)}(t) | t \in [0, T]\}$ with

$$\hat{P}_{0j}^{(\ell)}(t) = -\frac{(-1 + e^{-\hat{\alpha}_0^{(\ell)} t}) \hat{\alpha}_{0j}^{(\ell)}}{\hat{\alpha}_0^{(\ell)}} \quad (9)$$

for $j = 1, \dots, k$.

2.2 Random Right Censoring

According to Andersen et al. (2012), we model random right censoring by introducing the independent random variables $C^{(1)}$ and $C^{(2)}$ that denote the censoring times for group 1 and 2, respectively. We assume that these censoring times have a parametric distribution with density function $g^1(t, \psi^{(1)})$ for group 1 and $g^2(t, \psi^{(2)})$ for group 2, where $\psi^{(1)}, \psi^{(2)}$ are the parameters specifying the distributions. Furthermore, the censoring times are assumed to be independent of the survival times $T^{(1)}$ and $T^{(2)}$ to yield non-informative censoring.

For the samples $X_1^{(1)}, \dots, X_{n_1}^{(1)}$ and $X_1^{(2)}, \dots, X_{n_2}^{(2)}$, the corresponding censoring times are denoted by $C_1^{(1)}, \dots, C_{n_1}^{(1)}$ and $C_1^{(2)}, \dots, C_{n_2}^{(2)}$, respectively. Note that we do not observe the values $T_i^{(\ell)}$ or $C_i^{(\ell)}$ in the sample separately. We only observe the possibly censored survival times

$$\tilde{T}_i^{(\ell)} = \min(T_i^{(\ell)}, C_i^{(\ell)})$$

and the states $X_i^{(\ell)}(\tilde{T}_i^{(\ell)})$ of the patients at these time points. Then, the sample from group ℓ is given by

$$(\tilde{T}_1^{(\ell)}, X_1^{(\ell)}(\tilde{T}_1^{(\ell)})), \dots, (\tilde{T}_{n_\ell}^{(\ell)}, X_{n_\ell}^{(\ell)}(\tilde{T}_{n_\ell}^{(\ell)})),$$

and the contribution of an individual with $X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = j$ for $j = 1, \dots, k$ to the likelihood is given by

$$\mathbb{P}(T^{(\ell)} = \tilde{T}_i^{(\ell)}, X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = j) \cdot \mathbb{P}(C^{(\ell)} \geq \tilde{T}_i^{(\ell)}).$$

If we observe an individual with $X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = 0$, then it is censored and its contribution to the likelihood is given by

$$\mathbb{P}(T^{(\ell)} > \tilde{T}_i^{(\ell)}) \cdot \mathbb{P}(C^{(\ell)} = \tilde{T}_i^{(\ell)}).$$

This yields for the log-likelihood function for the group $\ell = 1, 2$

$$\begin{aligned} \log \mathcal{L}^{(\ell)}(\alpha^{(\ell)}, \psi^{(\ell)}) &= \sum_{i=1}^{n_\ell} \log(S^{(\ell)}(\tilde{T}_i^{(\ell)})) + \sum_{i=1}^{n_\ell} \sum_{j=1}^k I\{X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = j\} \log(\alpha_{0j}^{(\ell)}) \\ &\quad + \sum_{i=1}^{n_\ell} I\{X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = 0\} \log \left(\frac{g^\ell(\tilde{T}_i^{(\ell)}, \psi^{(\ell)})}{\mathbb{P}(C_i^{(\ell)} \geq \tilde{T}_i^{(\ell)})} \right) \\ &\quad + \sum_{i=1}^{n_\ell} \log(\mathbb{P}(C_i^{(\ell)} \geq \tilde{T}_i^{(\ell)})). \end{aligned} \quad (10)$$

Note that random right censoring does not influence the MLEs of the transition intensities, that is the MLEs are still given by (7). However, we can now additionally calculate the MLEs of the parameters of the censoring distribution from the log-likelihood function in (10) that is adapted to the random right censoring.

2.3 Similarity Test of Competing Risk Models Based on Transition Probabilities

We investigate the similarity of two competing risk models with transition probability functions $\{P_{0j}^{(1)}\}_{j=1,\dots,k}$ and $\{P_{0j}^{(2)}\}_{j=1,\dots,k}$, based on two respective samples, $X_1^{(1)}, \dots, X_{n_1}^{(1)}$ and $X_1^{(2)}, \dots, X_{n_2}^{(2)}$ over a time interval $[0, \tau]$ with fixed $\tau \in \mathbb{R}$. In the case of administrative censoring, the natural choice is $\tau = T$, where T is the censoring time. For random right censoring, τ should be chosen with respect to the research question. Due to the good interpretability of the maximum deviation, we define the similarity measure

$$d_\infty := \max_{j=1}^k \max_{t \in [0, \tau]} |P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t)| \quad (11)$$

and propose to test the hypotheses

$$H_0 : d_\infty \geq \varepsilon_\infty \quad \text{vs.} \quad H_1 : d_\infty < \varepsilon_\infty, \quad (12)$$

where $\varepsilon_\infty > 0$ is a pre-specified threshold defining when the models are considered as similar (for a more detailed discussion about the choice of ε_∞ see Remark 1(b)). Note that the alternative $H_1 : d_\infty < \varepsilon_\infty$ means that all transition probabilities of model 1 and 2 do not differ more than ε_∞ at any time. For these hypotheses, we define in Algorithm 1 and Algorithm 2 two tests corresponding to the different types of censoring discussed in Section 2.1 and 2.2, respectively.

Algorithm 1

Similarity Test via Constrained Parametric Bootstrap for Administrative Censoring

1. For $\ell = 1, 2$, calculate the MLEs $\hat{\alpha}^{(\ell)} = (\hat{\alpha}_{01}^{(\ell)}, \dots, \hat{\alpha}_{0k}^{(\ell)})^T$ of the transition intensities defined in (7) as well as the corresponding estimators of the transition probabilities

$$\hat{P}_{0j}^{(\ell)} = \{\hat{P}_{0j}^{(\ell)}(t) | t \in [0, \tau]\}$$

for $j = 1, \dots, k$ defined in (9). Then, calculate the estimator of the test statistic

$$\hat{d}_\infty = \max_{j=1}^k \max_{t \in [0, \tau]} |\hat{P}_{0j}^{(1)}(t) - \hat{P}_{0j}^{(2)}(t)|. \quad (13)$$

2. Define the following constrained estimators $\hat{\bar{\alpha}}^{(\ell)} = (\hat{\bar{\alpha}}_{01}^{(\ell)}, \dots, \hat{\bar{\alpha}}_{0k}^{(\ell)})$, $\ell = 1, 2$ for the transition intensities by

$$\hat{\bar{\alpha}}^{(\ell)} = \begin{cases} \hat{\alpha}^{(\ell)} & \text{if } \hat{d}_\infty \geq \varepsilon_\infty, \\ \bar{\alpha}^{(\ell)} & \text{if } \hat{d}_\infty < \varepsilon_\infty, \end{cases} \quad (14)$$

where $\bar{\alpha}^{(1)}$ and $\bar{\alpha}^{(2)}$ denote the MLEs of $\alpha^{(1)}$ and $\alpha^{(2)}$ under the constraint

$$\max_{j=1}^k \max_{t \in [0, \tau]} |P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t)| = \varepsilon_\infty. \quad (15)$$

That means, if $\hat{d}_\infty$ fulfills the null hypothesis, we choose the original MLEs and otherwise we optimize the log-likelihood function given in (6) with respect to the constraint above.

- 3.1. Generate bootstrap data

$$X_1^{*(1)}, \dots, X_{n_1}^{*(1)}, X_1^{*(2)}, \dots, X_{n_2}^{*(2)}$$

under the null hypothesis using the simulation approach described in Beyersmann et al. (2012) with the estimators $\hat{\bar{\alpha}}^{(1)}$ and $\hat{\bar{\alpha}}^{(2)}$, respectively. A single sample is generated by simulating the survival time of an individual based on the estimator of the all-cause hazard $\hat{\bar{\alpha}}_0^{(\ell)} = \sum_{j=1}^k \hat{\bar{\alpha}}_{0j}^{(\ell)}$ and, by deciding with a multinomial experiment with probabilities $\hat{\bar{\alpha}}_{0j}^{(\ell)} / \hat{\bar{\alpha}}_0^{(\ell)}$ what state the individual transitions to.

3.2. Analogous to (7), calculate the MLEs $\hat{\alpha}^{*(1)}$ and $\hat{\alpha}^{*(2)}$ based on the bootstrap data generated in Step 3.1. Then, for $\ell = 1, 2$, calculate the corresponding estimators of the transition probabilities

$$\hat{P}_{0j}^{*(\ell)} := \left\{ \hat{P}_{0j}^{*(\ell)}(t) \mid t \in [0, \tau] \right\} = \left\{ -\frac{(-1 + e^{-\hat{\alpha}_0^{*(\ell)} t}) \hat{\alpha}_{0j}^{*(\ell)}}{\hat{\alpha}_0^{*(\ell)}} \mid t \in [0, \tau] \right\},$$

where

$$\hat{\alpha}_0^{*(\ell)} = \sum_{j=1}^k \hat{\alpha}_{0j}^{*(\ell)}. \quad (16)$$

Furthermore, calculate the bootstrap statistic

$$\hat{d}_{\infty}^* = \max_{j=1}^k \max_{t \in [0, \tau]} \left| \hat{P}_{0j}^{*(1)}(t) - \hat{P}_{0j}^{*(2)}(t) \right|. \quad (17)$$

3.3. Repeat the steps 3.1. and 3.2. B times to generate B replicates of the test statistic, and denote by

$$\hat{d}_{\infty(1)}^* \leq \dots \leq \hat{d}_{\infty(B)}^*$$

the order statistic of these replicates.

4. Calculate the empirical α -quantile

$$\hat{q}_{\alpha}^* := \hat{d}_{\infty(\lfloor \alpha B \rfloor)}^* \quad (18)$$

that is an estimator of the true bootstrap α -quantile q_{α}^* of the distribution of $\hat{d}_{\infty}^*$ in (17).

5. Reject the null hypothesis in (12) if

$$\hat{d}_{\infty} < \hat{q}_{\alpha}^*.$$

Algorithm 2

Similarity Test via Constrained Parametric Bootstrap for Random Right Censoring

1. For $\ell = 1, 2$, calculate the MLEs $\hat{\alpha}^{(\ell)} = (\hat{\alpha}_{01}^{(\ell)}, \dots, \hat{\alpha}_{0k}^{(\ell)})^T$ defined in (7) as well as the MLEs $\hat{\psi}^{(\ell)}$ obtained by the optimization of the log-likelihood in (10) with respect to the parameters $\psi^{(\ell)}$. Then, calculate the corresponding plug-in estimators of the transition probabilities

$$\hat{P}_{0j}^{(\ell)} = \left\{ \hat{P}_{0j}^{(\ell)}(t) \mid t \in [0, \tau] \right\}$$

with the approach described in Feller Feller (1940) and the estimator of the test statistic $\hat{d}_{\infty}$ given in (13).

2. For $\ell = 1, 2$, we define the constrained estimators of the transition intensities and the censoring parameter as

$$(\hat{\alpha}^{(\ell)}, \hat{\psi}^{(\ell)}) = \begin{cases} (\hat{\alpha}^{(\ell)}, \hat{\psi}^{(\ell)}) & \text{if } \hat{d}_{\infty} \geq \varepsilon, \\ (\bar{\alpha}^{(\ell)}, \bar{\psi}^{(\ell)}) & \text{if } \hat{d}_{\infty} < \varepsilon, \end{cases} \quad (19)$$

where $(\bar{\alpha}^{(1)}, \bar{\psi}^{(1)})$ and $(\bar{\alpha}^{(2)}, \bar{\psi}^{(2)})$ denote the MLEs of $(\alpha^{(1)}, \psi^{(1)})$ and $(\alpha^{(2)}, \psi^{(2)})$ under the constraint (15). That is we optimize the log-likelihood function given in (10) with respect to the constraint (15).

3.1. We generate bootstrap data

$$X_1^{*(1)}, \dots, X_{n_1}^{*(1)}, X_1^{*(2)}, \dots, X_{n_2}^{*(2)}$$

under the null hypothesis with the procedure described in Beyersmann et al. Beyersmann et al. (2012) and with the transition intensities and censoring parameters $(\hat{\alpha}^{(1)}, \hat{\psi}^{(1)})$ and $(\hat{\alpha}^{(2)}, \hat{\psi}^{(2)})$, respectively.

- 3.2. Calculate the MLEs $\hat{\alpha}^{*(1)}, \hat{\alpha}^{*(2)}, \hat{\psi}^{*(1)}$ and $\hat{\psi}^{*(2)}$ based on the bootstrap data generated in Step 3.1. Then, for $\ell = 1, 2$, calculate the corresponding estimators of the transition probabilities denoted by

$$\hat{P}_{0j}^{*(\ell)} = \left\{ \hat{P}_{0j}^{*(\ell)}(t) \mid t \in [0, \tau] \right\}$$

by the approach in Feller (1940) and the bootstrap test statistic $\hat{d}_{\infty}^*$ defined in (17).

Steps 3.3., 4. and 5. are analogous to those in Algorithm 1.

Algorithm 3

Similarity Test via Constrained Parametric Bootstrap for Random Right Censoring and Test Statistic without Censoring

1. For $\ell = 1, 2$, calculate the MLEs $\hat{\alpha}^{(\ell)} = (\hat{\alpha}_{01}^{(\ell)}, \dots, \hat{\alpha}_{0k}^{(\ell)})^T$ defined in (7) as well as the MLEs $\hat{\psi}^{(\ell)}$ obtained by the optimization of the log-likelihood in (10) with respect to the parameters $\psi^{(\ell)}$. Then, calculate the corresponding plug-in estimators of the transition probabilities

$$\hat{P}_{0j}^{(\ell)} = \left\{ \hat{P}_{0j}^{(\ell)}(t) \mid t \in [0, \tau] \right\}$$

with the approach described in Feller (1940) and the estimator of the test statistic $\hat{d}_{\infty}$ given in (13).

2. For $\ell = 1, 2$, we define the constrained estimators of the transition intensities as

$$\hat{\hat{\alpha}}^{(\ell)} = \begin{cases} \hat{\alpha}^{(\ell)} & \text{if } \hat{d}_{\infty} \geq \varepsilon, \\ \bar{\alpha}^{(\ell)} & \text{if } \hat{d}_{\infty} < \varepsilon, \end{cases} \quad (20)$$

where $(\bar{\alpha}^{(1)}, \bar{\alpha}^{(2)})$ denotes the MLE of $(\alpha^{(1)}, \alpha^{(2)})$ under the constraint (15). That is we optimize the log-likelihood function given in (10) with respect to the constraint (15).

- 3.1. We generate bootstrap data

$$X_1^{*(1)}, \dots, X_{n_1}^{*(1)}, X_1^{*(2)}, \dots, X_{n_2}^{*(2)}$$

under the null hypothesis with the procedure described in Beyersmann et al. (2012) with the transition intensities $(\hat{\alpha}^{(1)}, \hat{\alpha}^{(2)})$ and censoring parameters $(\hat{\psi}^{(1)}, \hat{\psi}^{(2)})$.

- 3.2. Calculate the MLEs $\hat{\alpha}^{*(1)}$ and $\hat{\alpha}^{*(2)}$, based on the bootstrap data generated in Step 3.1. Then, for $\ell = 1, 2$, calculate the corresponding estimators of the transition probabilities denoted by

$$\hat{P}_{0j}^{*(\ell)} = \left\{ \hat{P}_{0j}^{*(\ell)}(t) \mid t \in [0, \tau] \right\}$$

by the approach in Feller (1940) and the bootstrap test statistic $\hat{d}_{\infty}^*$ defined in (17).

Steps 3.3., 4. and 5. are analogous to those in Algorithm 1.

Remark 1.

- (a) In Theorem 1 in the Appendix, we show that Algorithm 1 and 2 define valid test procedures for the hypotheses in (12) under administrative and random right censoring with exponentially distributed censoring times, respectively. Under the null hypothesis the test based on the transition probabilities keeps its nominal level asymptotically. A more precise statement can be made if the set

$$\mathcal{E} = \left\{ (u, i) \in [0, \tau] \times \{1, \dots, k\} \mid |P_{0i}^{(1)}(u) - P_{0i}^{(2)}(u)| = \max_{j=1}^k \max_{t \in [0, \tau]} |P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t)| \right\} \quad (21)$$

of all pairs of states and time points where the (absolute) difference between transition probabilities is maximal has only one element. In this case, the type I error converges to 0 inside the null hypothesis ($d_{\infty} > \varepsilon_{\infty}$) and to the nominal level α on the margin ($d_{\infty} = \varepsilon_{\infty}$) for increasing sizes $n_1, n_2 \rightarrow \infty$. Lastly, the test based on the transition probabilities detects the alternative ($d_{\infty} < \varepsilon_{\infty}$) with a probability converging to 1 with increasing sample size, which means that it is consistent.

- (b) An essential ingredient in testing the hypotheses (12) is the choice of the threshold ε_∞ , which has to be carefully discussed for each application. In situations, where it is difficult to fix the threshold in advance we can also define a threshold in a data driven way. To be precise, note that the statistic $\hat{d}_\infty^*$ in (17) depends on the threshold ε_∞ and in this remark we make this dependence explicit using the notation $\hat{d}_{\infty, \varepsilon_\infty}^*$. Note also that the hypotheses in (12) are nested. Recalling the definition of the bootstrap in Algorithm 1 and 2 it is easy to see that $\hat{d}_{\infty, \varepsilon_1}^* \leq \hat{d}_{\infty, \varepsilon_2}^*$ whenever $\varepsilon_1 \leq \varepsilon_2$. Consequently, we obtain for the corresponding quantiles in (18) the inequality $\hat{q}_{\alpha, \varepsilon_1}^* \leq \hat{q}_{\alpha, \varepsilon_2}^*$, and rejecting the null hypothesis in (12) by the test for $\varepsilon_\infty = \varepsilon_0$ also yields rejection of the null for all $\varepsilon_\infty > \varepsilon_0$. Therefore, by the sequential rejection principle, we may simultaneously test the hypotheses in (12) for different $\varepsilon_\infty > 0$ starting at a ε_∞ close to 0 and increasing ε_∞ to find the minimum value $\hat{\varepsilon}_\alpha$ for which H_0 is rejected for the first time. This value could be interpreted as a measure of evidence for similarity with a controlled type I error rate α .

Remark 2. In Section 3, we perform the test described in Algorithm 2 for exponentially distributed censoring times $C^{(1)}$ and $C^{(2)}$ with rate parameters $\psi^{(1)}, \psi^{(2)} > 0$, respectively. Recalling the notation from Section 2.2, it is easy to see that, in this special case, the maximum-likelihood estimator for the parameters $\psi^{(1)}, \psi^{(2)}$ in the censoring distribution are given by

$$\hat{\psi}^{(\ell)} = \frac{\sum_{i=1}^{n_\ell} I\{X_i^{(\ell)}(\tilde{T}_i^{(\ell)}) = 0\}}{\sum_{i=1}^{n_\ell} \tilde{T}_i^{(\ell)}} \quad (\ell = 1, 2). \quad (22)$$

In other words, the maximum likelihood estimator (MLE) of $\psi^{(\ell)}$ is defined as the number of censored individuals in group ℓ divided by the total observed survival time of all individuals in group ℓ .

3 Simulation study

In this section, we investigate the finite-sample properties of the two tests proposed in Algorithm 1 and 2 and compare them with a similarity test recently proposed by Möllenhoff et al. (2024), which is based on a similarity measure involving the transition intensities, that is

$$d_{int} := \max_{j=1, \dots, k} |\alpha_{0j}^{(1)} - \alpha_{0j}^{(2)}|. \quad (23)$$

The corresponding hypotheses are therefore defined by

$$H_0 : d_{int} \geq \varepsilon_{int} \text{ vs. } H_1 : d_{int} < \varepsilon_{int} \quad (24)$$

with a threshold $\varepsilon_{int} > 0$. Note that this threshold is not necessarily the same as the threshold ε_∞ used in the hypotheses (12) as the hypotheses in (24) are based on a different similarity measure. Therefore, for a reasonable comparison both thresholds have to be adjusted (see the discussion below). As the similarity measure in (11) utilizes the transition probabilities to quantify similarity of competing risks models, we are often referring to this method as the test based on transition probabilities. The similarity measure of Möllenhoff et al. (2024) defined in (23) uses transition intensities to quantify similarity. Thus, we call this method the test based on transition intensities. In the following comparison, we investigate, whether transition intensities or transition probabilities are better suited to capture the underlying similarity of competing risks models.

For both tests, we evaluate the performance for administrative censoring and exponential random right censoring. In the first setting, individuals who do not transition to any of the competing risks states during a fixed 90-day follow-up period are administratively censored.

Table 1: Thresholds for the similarity tests based on transition intensities (ε_{int}) and the tests based on transition probabilities (ε_{∞}) under different censoring scenarios

Censoring	ε_{int}	ε_{∞}
adm	0.0015	0.11805
Exp(0.002)	0.0015	0.10849
Exp(0.005)	0.0015	0.0960
Exp(0.01)	0.0015	0.0794

Thus, we choose the time interval with end point $\tau = 90$ to specify the similarity measure defined in (11). In the second setting, the censoring times $C^{(1)}$, $C^{(2)}$ are exponentially distributed with rate parameters $\psi^{(1)}, \psi^{(2)} > 0$, respectively. To study the effect of different censoring rates, we consider the parameters $\psi^{(1)} = \psi^{(2)} \in \{0.002, 0.005, 0.01\}$ in the censoring distribution. In order to guarantee the comparability of the simulation results for administrative and random right censoring, we opt for $\tau = 90$ in this setting as well.

As the transition intensities uniquely determine a competing risks model, we define our simulation scenarios in terms of the transition intensities. To compare the tests based on transition intensities and probabilities, we define a simulation scenario which shall correspond to the margin of the hypotheses (12), that is $d_{\infty} = \varepsilon_{\infty}$. More specifically, we consider the transition intensities

$$\begin{aligned} (\alpha_{01}^{(1)}, \alpha_{02}^{(1)}, \alpha_{03}^{(1)}) &= (0.0023, 0.0011, 0.0004), \\ (\alpha_{01}^{(2)}, \alpha_{02}^{(2)}, \alpha_{03}^{(2)}) &= (0.0008, 0.0026, 0.0019) \end{aligned} \tag{25}$$

for group 1 and 2, respectively. For these transition intensities, the censoring rates $\psi^{(1)} = \psi^{(2)} \in \{0.002, 0.005, 0.01\}$ result in around 34% to 73% of censored individuals in group 1 and in around 27% to 65% of censored individuals in group 2. For the administrative censoring, we get 71% of censored individuals in group 1 and 62% in group 2. The similarity measure d_{int} , defined in (23), is equal to 0.0015 for the transition intensities in (25). Since d_{int} is independent of the censoring distribution, this equality holds across all censoring mechanisms above, and we choose the threshold ε_{int} such that the simulation scenario in (25) corresponds to the margin for the hypotheses in (12), that is

$$\varepsilon_{int} = d_{int} = 0.0015 \tag{26}$$

(this choice does not depend on the censoring setting). In order to make the results of both tests comparable we now calculate the value d_{∞} for this scenario and choose the threshold ε_{∞} for the test based on transition probabilities, such that $\varepsilon_{\infty} = d_{\infty}$. However, as can be seen by (4) and (??), the value d_{∞} depends on the type of censoring, and therefore its calculation is a little more difficult. For administrative censoring we use (4) and (11) to calculate d_{∞} from the transition intensities given in (25) and obtain $d_{\infty} = 0.11805$. In this case, we define the threshold for the hypotheses (24) by $\varepsilon_{\infty} = d_{\infty} = 0.11805$. In the case of exponential random right censoring, we utilize formula (??) to compute the transition probabilities from the transition intensities given in (25) and the respective rate parameter to obtain d_{∞} . Then, we again define $\varepsilon_{\infty} = d_{\infty}$. The values of ε_{∞} for the different rate parameters can be found in Table 1. For example, we obtain $\varepsilon_{\infty} = 0.096$ in the setting of random right censoring with rate parameter 0.005. By these choices, we guarantee that the scenario (25) corresponds to the margin for the hypotheses (12) and (24) such that the tests based on transition intensities and probabilities are comparable.

To investigate the performance of the tests under administrative censoring, we generate data on the margin by simulating healthcare pathways from the transition intensities in (25) with the algorithm described in Beyersmann et al. (2012). The observed survival times are given by the minimum of the generated survival time and $\tau = 90$. To simulate data inside the null hypothesis,

Table 2: Transition intensities $\alpha^{(1)}$ and $\alpha^{(2)}$ to simulate healthcare pathways inside the null hypothesis, on the margin and under the alternative hypothesis.

Scenario	$\alpha_{01}^{(1)}$	$\alpha_{02}^{(1)}$	$\alpha_{03}^{(1)}$	$\alpha_{01}^{(2)}$	$\alpha_{02}^{(2)}$	$\alpha_{03}^{(2)}$
Null	0.0028	0.0011	0.0004	0.0008	0.0028	0.0019
Margin	0.0023	0.0011	0.0004	0.0008	0.0026	0.0019
Alternative 1	0.0018	0.0011	0.0004	0.0008	0.0021	0.0014
Alternative 2	0.0013	0.0011	0.0004	0.0008	0.0016	0.0014
Alternative 3	0.0010	0.0011	0.0004	0.0008	0.0013	0.0009
Alternative 4	0.0009	0.0011	0.0004	0.0008	0.0012	0.0007
Alternative 5	0.0009	0.0011	0.0004	0.0008	0.0012	0.0005

we generate healthcare pathways from transition intensities with a similarity measure d_∞ being larger than the chosen thresholds. To simulate data under the alternative hypothesis, we choose transition intensities with a similarity measure d_∞ being smaller than the chosen thresholds. Overall, we design one scenario inside the null and five scenarios under the alternative, being progressively further away from the margin. These scenarios are described in Table 2, together with the initial margin case.

To evaluate the performance of the similarity test based on transition probabilities for random right censoring (Algorithm 2), we choose the same scenarios as described in Table 2. Additionally to the event times, we generate censoring times from the $\text{Exp}(\psi^{(\ell)})$ -distribution. The observed survival time is the minimum of survival time and censoring time.

We investigate the performance of our method and compare it to the method of Möllenhoff et al. (2024) for a wide variety of sample sizes, i.e. $(n_1, n_2) \in \{(50, 50), (100, 100), (200, 200), (300, 300), (250, 450), (500, 500)\}$. The choice of $(n_1, n_2) = (250, 450)$ is close to the observed sample size in the real data example in Section 4 and enables us to analyze the effect of unbalanced sample sizes. Furthermore, the first four sample sizes $(50, 50)$, $(100, 100)$, $(200, 200)$, $(300, 300)$ correspond to settings where we observe even less competing risks events than in the real data set.

The percentage of censored individuals changes depending on the scenario from Table 2. Hereby, the proportion of censored individuals is the smallest for the null scenario, and grows from alternative 1 to alternative 5. In the case of $\text{Exp}(0.01)$ censoring, around 70% of patients of group 1 are censored in the null scenario, while around 81% are censored for Alternative 5.

The performance of the test based on transition probabilities is assessed in terms of type I error and power that are documented in Tables 3 and 4. The first entry in each table represents the empirical rejection probabilities of the testing procedure presented in this article (Algorithm 1 and 2). The number in brackets corresponds to the empirical rejection probabilities of the test suggested in Möllenhoff et al. (2024). All tests are performed with a nominal level of $\alpha = 0.05$ with $B = 500$ bootstrap repetitions and 1000 simulation runs are used for each scenario to calculate the empirical rejection probabilities.

3.1 Type I error

Table 3 summarizes the simulated type I error under the null hypothesis. The empirical results confirm the theoretical statements in Theorem 1. First note that the set $\mathcal{E}$ defined in (21) has cardinality 1 in all cases under consideration, and we expect (by the discussion in Remark 1(a)) that for an increasing sample size the empirical rejection probabilities inside the null hypothesis should converge to 0. This is confirmed by our simulation study. The empirical type I error of the test based on the transition probabilities inside the null hypothesis varies between 0% and 0.8%. That means it is already close to 0 for the small sample size $(50, 50)$ across all censoring

mechanisms and indicates that for these scenarios there is no evidence for similarity.

For a discussion of the properties of the test at the margin, note that by the discussion in Remark 1(a) we expect that for an increasing sample size the empirical rejection probabilities should approximate the nominal level $\alpha = 5\%$ for all considered cases (the precise statement can be found in Theorem 1(2) in the supplement). The type I error of the new test varies between 1% to 4%. Increasing the sample size brings the type I error closer to the nominal 5% level, confirming the theoretical statement. For instance, the type I error on the margin for Exp(0.002) censoring increases from 0.018 for a sample size of (50, 50) to 0.041 for a sample size of (500, 500). Also, for a fixed sample size, higher exponential censoring rates generally reduce the type I error at the margin. For example, at sample size (200, 200), the proportion of rejection decreases from 0.016 for a censoring rate of $\psi^{(\ell)} = 0.002$, to 0.014 for $\psi^{(\ell)} = 0.005$, and further to 0.007 for $\psi^{(\ell)} = 0.01$.

For the test based on the transition intensities proposed in Möllenhoff et al. (2024), the type I error inside the null hypothesis is smaller than 0.4% across all sample sizes and censoring mechanisms. For the margin scenario, the type I error is smaller than 1% and, therefore, always significantly smaller than the respective error of our test. This suggests that the test based on transition intensities is more conservative than the test based on transition probabilities.

3.2 Power

The simulated power of the test based on transition probabilities is displayed in Table 4. For a fixed sample size and a fixed censoring mechanism, power increases from alternative 1 to alternative 5. For example, if $(n_1, n_2) = (100, 100)$ and Exp(0.002) censoring is considered, power rises from 0.246 for alternative 1 to 0.996 for alternatives 4 and 5. As the similarity measure d_∞ for the transition probabilities of the two groups in Table 2 increases, the test can detect these alternatives better. The numbers in brackets represent the corresponding empirical rejection probabilities of the similarity test based on transition intensities (Möllenhoff et al. (2024)). We observe that the new test based on transition probabilities has more power in nearly all scenarios under consideration and a more detailed comparison is given at the end of this section.

For a fixed sample size, a larger censoring rate in exponential random right censoring yields to a loss of power (for the test based on transition intensities as well as for the test based on transition probabilities). This is expected as the number of observed transitions into one of the competing risks states decreases with increasing censoring rate. For example, at $(n_1, n_2) = (300, 300)$, the power of the test based on the transition probabilities for alternative 2 drops from 0.928 at rate 0.002 to 0.612 at rate 0.01. In this example, the power for administrative censoring is 0.652. It can be observed across all sample sizes and alternatives that the power in the administrative censored case is slightly larger than in the Exp(0.01) censored case. This can be explained by the fact that the proportion of administratively censored individuals lies always just below the censored proportion for Exp(0.01) censoring.

Figure 1 displays these trends for the test based on transition probabilities for equal group sizes $n_1 = n_2 = 200$. Here, the difference in power between the censoring mechanisms is largest for alternative 1 and 2. For alternative 3 to 5 the test has power close to 1 across all censoring mechanisms, even though the power for mechanisms yielding smaller proportions of censored individuals tends to 1 faster.

We observe that the power increases with the sample size. Under the strong alternatives 3 to 5, the power is essentially 1 for sample sizes larger than $(n_1, n_2) = (300, 300)$, irrespective of the censoring mechanism. This lines up with the theoretical result from Theorem 1(3) stating that the power of the tests described in Algorithm 1 and 2 converges to 1 for $n_1, n_2 \rightarrow \infty$. The power for the sample size $(n_1, n_2) = (250, 450)$ being larger than for $(n_1, n_2) = (300, 300)$ in every case, suggests that the test also performs well in settings with unbalanced group sizes.

Figure 2 illustrates how the power of the test based on the transition probabilities improves

Table 3: Empirical rejection probabilities of the similarity test based on transition probabilities under administrative (Algorithm 1) and exponential random right censoring (Algorithm 2) on the margin (Margin) and inside the null hypothesis (Null) for different censoring mechanisms and different sample sizes. The numbers in brackets correspond to the empirical rejection probabilities of the similarity test based on transition intensities ((Möllenhoff et al., 2024)). The nominal level is $\alpha = 0.05$.

Sample size	Censoring	Margin	Null
(50, 50)	Exp(0.002)	0.018 (0.004)	0.001 (0.000)
	Exp(0.005)	0.014 (0.005)	0.003 (0.000)
	Exp(0.01)	0.017 (0.010)	0.008 (0.002)
	adm	0.012 (0.005)	0.004 (0.004)
(100, 100)	Exp(0.002)	0.018 (0.005)	0.002 (0.000)
	Exp(0.005)	0.016 (0.006)	0.003 (0.001)
	Exp(0.01)	0.014 (0.004)	0.005 (0.002)
	adm	0.016 (0.006)	0.002 (0.001)
(200, 200)	Exp(0.002)	0.016 (0.002)	0.001 (0.001)
	Exp(0.005)	0.014 (0.003)	0.002 (0.000)
	Exp(0.01)	0.007 (0.001)	0.002 (0.000)
	adm	0.013 (0.005)	0.003 (0.000)
(300, 300)	Exp(0.002)	0.027 (0.005)	0.000 (0.000)
	Exp(0.005)	0.028 (0.003)	0.000 (0.000)
	Exp(0.01)	0.021 (0.004)	0.001 (0.000)
	adm	0.014 (0.001)	0.004 (0.000)
(250, 450)	Exp(0.002)	0.032 (0.005)	0.000 (0.000)
	Exp(0.005)	0.031 (0.002)	0.000 (0.000)
	Exp(0.01)	0.021 (0.003)	0.003 (0.000)
	adm	0.029 (0.003)	0.004 (0.000)
(500, 500)	Exp(0.002)	0.041 (0.004)	0.000 (0.000)
	Exp(0.005)	0.030 (0.003)	0.001 (0.000)
	Exp(0.01)	0.021 (0.000)	0.000 (0.000)
	adm	0.020 (0.000)	0.004 (0.000)

with increasing sample size for the two alternatives 2 and 3. For both alternatives, lower proportions of censored individuals yield higher power. In alternative 3, the gap narrows with growing sample size, as nearly full power is already achieved for 300 observations per group for all censoring mechanisms. Under alternative 2, the test requires larger sample sizes to attain comparable power levels. Again, the power curves with respect to administrative censoring and Exp(0.01) censoring are similar, since these censoring mechanisms yield similar proportions of censored patients.

We conclude this section with a more detailed comparison of the test based on transition probabilities and the test based on transition intensities proposed in Möllenhoff et al. (2024). As already mentioned, the power of the similarity test based on transition probabilities consistently exceeds that of the test based on transition intensities across nearly all scenarios, censoring mechanisms and sample sizes. For sample size $n_1 = n_2 = 200$ and Exp(0.002) censoring, the gain of power by using the test based on transition probabilities is 0.317 for alternative 1. Figure 3 provides a direct comparison of the two tests for all scenarios from Table 2 and all censoring scenarios for the sample size $(n_1, n_2) = (200, 200)$. The gain of power is most pronounced under the intermediate alternatives 1 to 3. Under the stronger alternatives 4 and 5, both tests have

Table 4: Empirical rejection probabilities of the similarity test based on transition probabilities for administrative (Algorithm 1) and random censoring (Algorithm 2) under different alternatives and for different censoring mechanisms and different sample sizes. The numbers in brackets correspond to the empirical rejection probabilities of the test based on transition intensities ((Möllenhoff et al., 2024)). The nominal level is $\alpha = 0.05$.

Sample size pair	Censoring mechanism	Alt1	Alt2	Alt3	Alt4	Alt5
(50, 50)	Exp(0.002)	0.107 (0.061)	0.289 (0.195)	0.775 (0.671)	0.848 (0.787)	0.869 (0.808)
	Exp(0.005)	0.069 (0.038)	0.214 (0.147)	0.519 (0.431)	0.617 (0.534)	0.610 (0.535)
	Exp(0.01)	0.069 (0.041)	0.124 (0.085)	0.275 (0.221)	0.355 (0.298)	0.303 (0.261)
	adm	0.075 (0.049)	0.148 (0.108)	0.309 (0.252)	0.374 (0.311)	0.400 (0.350)
(100, 100)	Exp(0.002)	0.246 (0.121)	0.542 (0.398)	0.979 (0.951)	0.996 (0.989)	0.996 (0.986)
	Exp(0.005)	0.139 (0.102)	0.337 (0.239)	0.855 (0.783)	0.929 (0.880)	0.929 (0.896)
	Exp(0.01)	0.089 (0.100)	0.234 (0.157)	0.578 (0.485)	0.655 (0.583)	0.658 (0.600)
	adm	0.105 (0.059)	0.254 (0.168)	0.630 (0.533)	0.740 (0.662)	0.764 (0.699)
(200, 200)	Exp(0.002)	0.517 (0.200)	0.828 (0.698)	0.999 (0.995)	1.000 (1.000)	1.000 (1.000)
	Exp(0.005)	0.312 (0.168)	0.689 (0.530)	0.996 (0.983)	0.996 (0.993)	1.000 (0.999)
	Exp(0.01)	0.204 (0.134)	0.459 (0.328)	0.905 (0.855)	0.939 (0.919)	0.942 (0.926)
	adm	0.193 (0.106)	0.497 (0.358)	0.927 (0.880)	0.964 (0.946)	0.977 (0.959)
(300, 300)	Exp(0.002)	0.731 (0.721)	0.928 (0.963)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
	Exp(0.005)	0.511 (0.554)	0.815 (0.873)	0.999 (0.998)	1.000 (1.000)	1.000 (1.000)
	Exp(0.01)	0.314 (0.282)	0.612 (0.535)	0.983 (0.977)	0.988 (0.982)	0.986 (0.985)
	adm	0.344 (0.250)	0.652 (0.614)	0.995 (0.991)	0.999 (0.998)	0.999 (0.999)
(250, 450)	Exp(0.002)	0.756 (0.520)	0.980 (0.815)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
	Exp(0.005)	0.551 (0.310)	0.908 (0.669)	1.000 (0.997)	1.000 (1.000)	1.000 (1.000)
	Exp(0.01)	0.349 (0.193)	0.748 (0.481)	0.987 (0.964)	0.987 (0.985)	0.985 (0.982)
	adm	0.422 (0.181)	0.792 (0.500)	0.995 (0.980)	1.000 (0.997)	0.999 (0.998)
(500, 500)	Exp(0.002)	0.902 (0.795)	0.990 (0.923)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
	Exp(0.005)	0.744 (0.572)	0.954 (0.896)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
	Exp(0.01)	0.509 (0.385)	0.843 (0.798)	0.999 (0.983)	0.998 (0.987)	0.998 (0.985)
	adm	0.570 (0.347)	0.845 (0.685)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)

nearly full power, though our test converges more rapidly. The relative benefit of the test based on transition probabilities becomes more evident in this example as the proportion of censored individuals decreases. These results underline that the use of transition probabilities in order to define similarity of competing risks models yields a more powerful similarity test.

4 Application Example: Healthcare Pathways of Prostate Cancer Patients Involving Surgery

In our application example, we analyze routine inpatient data of prostate cancer patients treated at the Department of Urology, University Medical Center Freiburg, between 1 January 2015 and 1 February 2021. All patients underwent open radical prostatectomy (ORP), involving removal of the prostate and seminal vesicles. The dataset comprises two groups: those who received in-house MRI-based fusion biopsy (FB) before surgery and those who did not. We aim to assess whether post-surgical pathways, specifically regarding short-term hospital readmission, are similar between these groups to evaluate whether data can be pooled. In total, 213 patients (31%) had in-house FB, while 482 (69%) did not. Prostate cancer is among the most common reasons

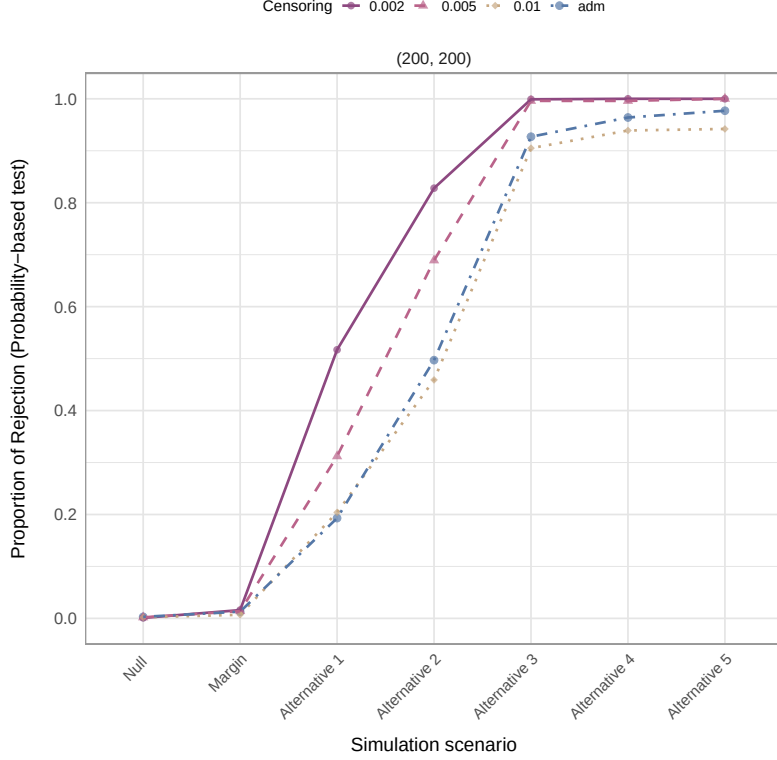


Figure 1: Empirical rejection probabilities of the similarity test based on transition probabilities for the scenarios in Table 2 ($n_1 = n_2 = 200$). The different lines represent the power for different censoring mechanisms.

for inpatient admission, and ORP is a standard treatment. While FB is the current diagnostic standard, not all patients undergo it before surgery, depending on the referring urologist’s practice. Hospital readmissions were categorized into three groups according to their main ICD-10 diagnosis, “I89.8: Other specified noninfective disorders of lymphatic vessels and lymph nodes”, “C61: Malignant neoplasm of prostate”, and a combined category for “any other diagnosis”. As the most common postoperative complications of ORP develop within a few weeks after surgery, we consider administrative censoring after 90 days in our analysis. Within this period, readmissions occurred for ICD-10 I89.8 (Model 1: $n = 17$ [8%], Model 2: $n = 29$ [6%]), ICD-10 C61 (Model 1: $n = 18$ [8%], Model 2: $n = 60$ [12%]), and “any other diagnosis” (Model 1: $n = 6$ [3%], Model 2: $n = 31$ [6%]). Readmissions occurring within the 90-day period were modeled using a competing risks modeling framework, with the initial states corresponding to “ORP with prior in-house FB” and “ORP without prior in-house FB” for the two groups, respectively. The absorbing states are defined by the three readmission categories. In this context, focusing on transition probabilities is more relevant than on intensities, as probabilities directly describe the likelihood of readmission within a clinically meaningful time frame. They are easier to interpret and compare across groups, whereas intensities reflect instantaneous event rates that are less intuitive for evaluating overall pathway similarity. Using the plug-in estimators described in Section 2, we first examine the estimated transition probabilities at selected landmark times. Table 5 displays the landmark risks at clinically relevant times (30, 60, and 90 days) for each event and group, together with the absolute differences and corresponding 95% bootstrap confidence intervals for the absolute differences. The confidence intervals were obtained using a parametric bootstrap based on the fitted exponential model with administrative censoring at 90 days. Across all event types, the absolute differences increase over time. The largest discrepancy

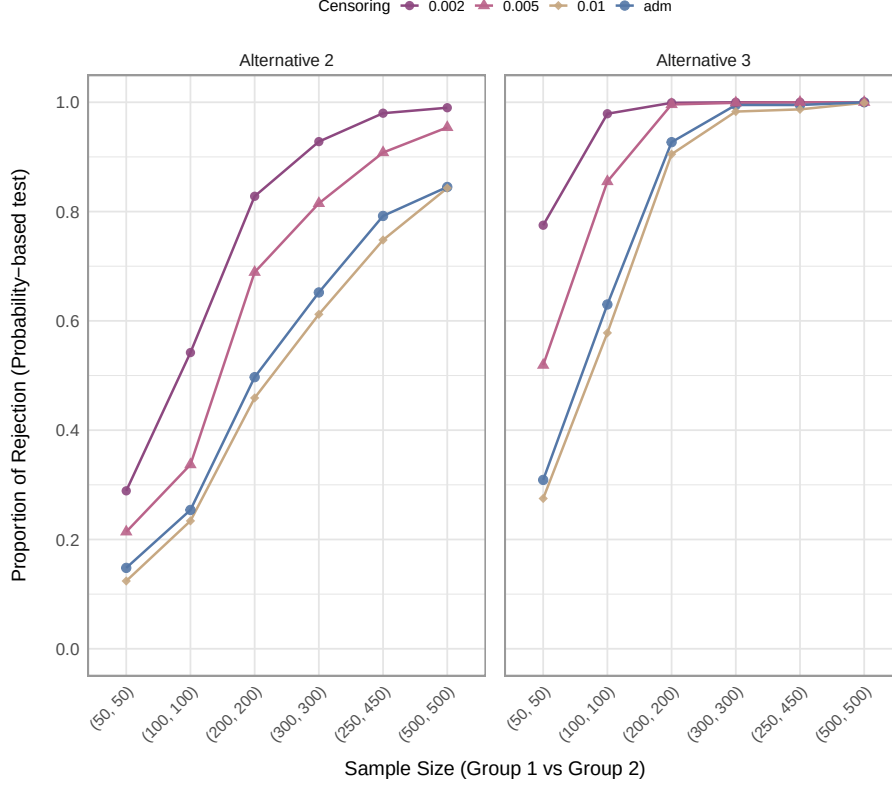


Figure 2: Empirical rejection probabilities of the similarity test based on transition probabilities under alternative 2 (left) and alternative 3 (right), in dependence of the group sample sizes. The lines represent the power for different censoring mechanisms.

is observed for readmissions due to ICD-10 C61, where the absolute difference reaches 0.0494 (95% CI: 0.0067–0.0968) at 90 days. Differences for ICD-10 I89.8 and for other diagnoses are overall smaller. These observed magnitudes provide a descriptive reference for interpreting the similarity thresholds considered in the formal testing procedure below.

To assess the similarity of healthcare pathways between the two groups, we apply the method described in Algorithm 1. As shown in Section 3, this approach performs well, outperforming a test based directly on transition intensities. We compute p-values for a range of similarity thresholds to determine at which values the two populations can be considered to have comparable readmission patterns. If similarity holds, pooling the two groups would be justified and would substantially increase the sample size for subsequent analyses. Table 6 reports the p-values from the test based on transition probabilities for ten thresholds. For thresholds between 0.05 and 0.08, the p-values exceed the nominal level of $\alpha = 0.05$, so the null hypothesis cannot be rejected. For thresholds of 0.087 and higher, the p-values fall below this level, indicating similarity between the two models. Thus, if a maximal difference in transition probabilities of about 0.087 is acceptable, we may claim similarity of both models, i.e., no difference regarding hospital readmission rates whether in-house FB has been performed or not.

5 Discussion

In this article, we develop a novel parametric bootstrap test to evaluate the similarity of two competing risks models with constant transition intensities. In contrast to previous work on similarity testing, which assesses the similarity of competing risks models in terms of transition

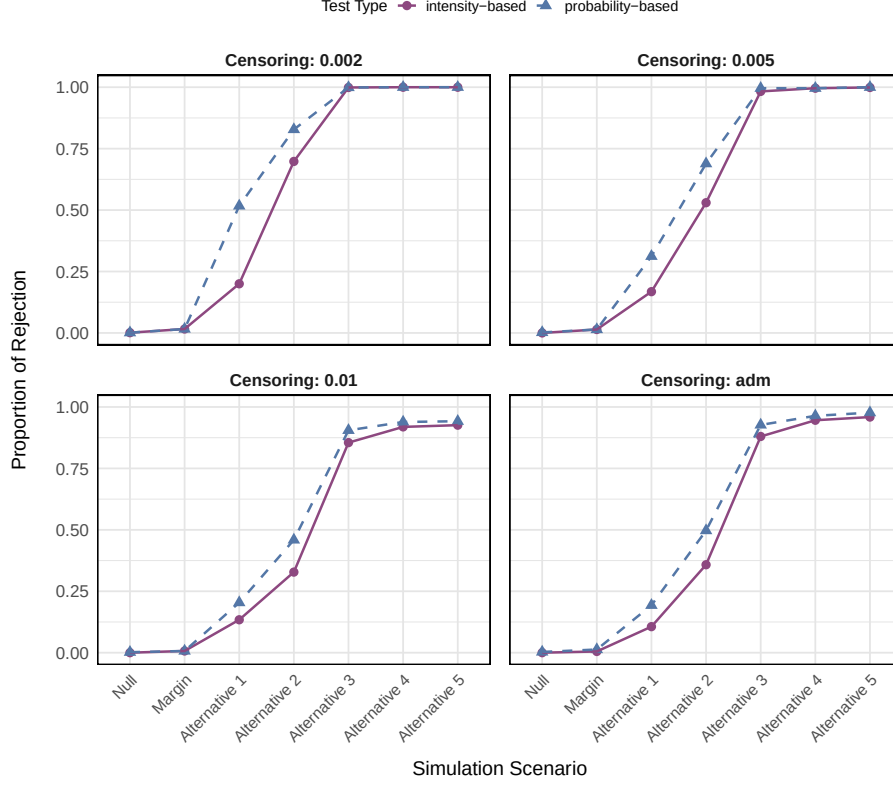


Figure 3: Empirical rejection probabilities of the similarity test based on transition probabilities and the test based on transition intensities under the scenarios from Table 2. The sample size is $(n_1, n_2) = (200, 200)$ and four censoring scenarios are considered.

intensities (Binder et al., 2022; Möllenhoff et al., 2024), our approach measures similarity between transition probabilities. This makes the results of the test easier to interpret in a medical context, where outcomes are typically assessed in terms of cumulative risks rather than instantaneous event rates. For the sake of comparability with the approach of Möllenhoff et al. (2024), we opted for a similarity measure based on the supremum distance. The choice of the similarity measure ultimately depends on the research question and exploring alternative distances remains an important branch of future research.

The performance of the proposed probability-based similarity test was evaluated through extensive simulation studies across a wide range of settings, including varying sample sizes, censoring rates, and degrees of model dissimilarity. This demonstrated that our proposed test reliably maintains the α level and yields high power across both considered censoring mechanisms, administrative censoring and exponential random right censoring. Furthermore, we applied the test proposed by Möllenhoff et al. (2024) under identical simulation settings and showed that the test based on transition probabilities consistently has higher power than the one of Möllenhoff et al. (2024).

This gain in power comes at the cost of added complexity in establishing theoretical results for similarity measures based on transition probabilities. The main reason for this is that discussing the differentiability of the function mapping the transition intensities onto the transition probabilities, needed to establish the asymptotic distribution of the test statistic, is very extensive for non-constant transition intensities. Consequently, the test proposed in this article is limited to models with constant transition intensities, despite the previous work by Möllenhoff et al. (2024) extending their method to models with parametric transition intensities. The assumption of constant transition intensities provides a controlled setting to evaluate methodological differences

Table 5: Estimated transition probabilities until 30, 60, and 90 days, respectively, for each event and group, including absolute differences and bootstrap 95% CIs.

Event	Time (days)	$\hat{P}_{0j}^{(1)}(t)$	$\hat{P}_{0j}^{(2)}(t)$	$ \hat{P}^{(1)} - \hat{P}^{(2)} $	95% CI for difference
ICD-10 I89.8	30	0.0297	0.0234	0.0063	[0.0005, 0.0227]
ICD-10 I89.8	60	0.0574	0.0447	0.0127	[0.0010, 0.0434]
ICD-10 I89.8	90	0.0833	0.0641	0.0191	[0.0013, 0.0626]
ICD-10 C61	30	0.0297	0.0484	0.0187	[0.0031, 0.0366]
ICD-10 C61	60	0.0574	0.0925	0.0351	[0.0053, 0.0688]
ICD-10 C61	90	0.0833	0.132	0.0494	[0.0067, 0.0968]
Any other diagnosis	30	0.0070	0.0169	0.0100	[0.0010, 0.0196]
Any other diagnosis	60	0.0135	0.0324	0.0189	[0.0019, 0.0374]
Any other diagnosis	90	0.0196	0.0464	0.0268	[0.0028, 0.0531]

Table 6: P-values of the similarity test based on transition probabilities described in Algorithm 1 for the application example above considering different thresholds ϵ . The bold numbers mark p-values below the nominal level of $\alpha = 0.05$.

Threshold ϵ	0.05	0.06	0.07	0.08	0.087	0.09	0.10	0.11	0.12	0.13
P-value	0.362	0.268	0.16	0.09	0.044	0.034	0.006	0.002	0.002	0.000

between intensity- and probability-based similarity measures. Because transition probabilities can be derived analytically from constant hazards, this setup isolates the contribution of the similarity measure itself from that of additional model flexibility. For the constant transition intensity setting, we proved that our presented testing procedure is an asymptotic level- α tests and consistent for administrative and exponential random right censoring. Overcoming the associated mathematical challenges and extending our method to competing risks models with parametric transition intensities is a central objective for future research. Additionally, broadening the framework to include more general multistate models with parametric transition intensities is also envisioned.

We may further note here that potential model misspecification always is an important issue when it comes to parametric modeling. While we do not investigate robustness of our approach to model misspecification in this manuscript, Möllenhoff et al. (2024) studied this issue extensively in dedicated simulations and reported that moderate misspecification had only minor impact, whereas stronger misspecification could lead to either mild type I error inflation or conservative behavior with reduced power. Given that our approach uses the same model class for event dynamics, we expect a qualitatively similar pattern.

Finally, we illustrated the practical utility of our method through a real-world application involving prostate cancer patients who underwent radical prostatectomy with or without prior in-house MRI-targeted fusion biopsy. The analysis focused on comparing the 90-day hospital readmission patterns between the two groups across multiple causes. By applying our test based on transition probabilities, we were able to determine threshold values below which the global null hypothesis of dissimilarity could be rejected. The chosen thresholds were informed by simulation results and clinical considerations, reflecting what would be regarded as a negligible difference in readmission risk. The procedure provides formal statistical evidence for deciding

on similarity between different patient groups. Given the lack of established clinical guidelines for selecting similarity thresholds, further methodological and application-focused research on threshold specification is warranted.

In this paper we were particularly interested in data analysis for small sample sizes, where the use of nonparametric methods is less appropriate. However an interesting direction of future research is the development of nonparametric methodology for the similarity problems considered in this paper, which can be used without specific model assumptions if a sufficient amount of data is available. As the cumulative intensities and transition probabilities can be estimated nonparametrically via the Nelson-Aalen and Aalen-Johansen estimator, respectively, (see, for example, Andersen et al., 2012), we expect that corresponding methodology can be developed using similar techniques as in Section 2.4 of Dette and Neumeier (2001).

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Authors’ Contributions

ZL developed the theoretical framework, methodology, designed the algorithms, the simulation study and wrote the discussion and the interpretive text linking simulations and theoretical results.

MF implemented the simulation code, generated the figures and tables and wrote the description of the simulation results, the introduction and the application example.

NB and HD provided supervision and guidance throughout the project.

Data Availability Statement

The code used in the simulation study (Section 3) and a tutorial on how to use it can be found at <https://github.com/MaryamFarhadizadeh/similarity-test-transition-probability>.

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Appendix

The following result states that Algorithm 1 and 2 define valid tests for the hypotheses in (12) under administrative and random censoring, respectively. Note that the result is stated with respect to the true bootstrap quantile q_α^* as the estimation error of $\hat{q}_\alpha^*$ can be made arbitrarily small by increasing the number B of bootstrap iterations.

Theorem 1. *Let $X^{(1)}$ and $X^{(2)}$ be two independent competing risks models as defined in Section 2.1 or Section 2.2 with transition intensities $\alpha_{0j}^{(\ell)} > 0$ for $\ell = 1, 2$. Assume that it holds for $n := n_1 + n_2$ that*

$$\lim_{n_1, n_2 \rightarrow \infty} \frac{n}{n_1} = c \in (1, \infty).$$

Algorithm 1 and Algorithm 2 define a consistent and asymptotic level- α test for the hypotheses (12) in the case of administrative censoring and exponential random right censoring, respectively. More precisely, we have

- (1) *Let F_Z denote the distribution function of the random variable Z defined in equation (28) below and assume that F_Z is continuous at its α -quantile $q_{\alpha, Z}$. If the null hypothesis H_0 in (12) holds and if $q_{\alpha, Z} < 0$, then,*

$$\limsup_{n_1, n_2 \rightarrow \infty} \mathbb{P}(\hat{d}_\infty < q_\alpha^*) \leq \alpha.$$

- (2) *If the null hypothesis H_0 in (12) holds and if the set $\mathcal{E}$ defined in (21) consists of one point, it follows for any $\alpha \in (0, 0.5)$ that*

$$\lim_{n_1, n_2 \rightarrow \infty} \mathbb{P}(\hat{d}_\infty < q_\alpha^*) = \begin{cases} 0 & \text{if } d_\infty > \varepsilon_\infty, \\ \alpha & \text{if } d_\infty = \varepsilon_\infty. \end{cases}$$

(3) If the alternative hypothesis H_1 in (12) holds, it follows for any $\alpha \in (0, 0.5)$ that

$$\lim_{n_1, n_2 \rightarrow \infty} \mathbb{P}(\hat{d}_\infty < q_\alpha^*) = 1.$$

Proof of Theorem 1. The proof is given for the case of administrative censoring, but can easily be adapted to the case of exponential random right censoring. Recall the definitions of the transition intensities $\alpha^{(1)}, \alpha^{(2)} \in \mathbb{R}^k$ in (3) and of the transition probabilities $P_{0j}^{(1)}, P_{0j}^{(2)}$ in (4). Furthermore, recall the definition of the MLEs $\hat{\alpha}^{(1)}, \hat{\alpha}^{(2)}$ of the transition intensities given in (7) as well as of the estimators of the transition probabilities $\hat{P}_{0j}^{(1)}, \hat{P}_{0j}^{(2)}$ in (9). By Theorem 6.1 in Albert Albert (1962) the vector

$$\sqrt{n}((\hat{\alpha}^{(1)} - \alpha^{(1)})^T, (\hat{\alpha}^{(2)} - \alpha^{(2)})^T)^T$$

converges weakly to a multivariate normal distribution with mean 0 and diagonal covariance matrix. Define the set $\mathcal{M} := [0, \tau] \times \{1, \dots, k\}$ and let $\ell^\infty(\mathcal{M})$ denote the set of bounded functions $g : \mathcal{M} \rightarrow \mathbb{R}$. The function mapping the transition intensities to the respective transition probability function $t \rightarrow P_{0j}(t)$ is, based on (4), given by

$$P_{0j} : \begin{cases} \mathbb{R}^k \rightarrow \ell^\infty(\mathcal{M}) \\ \alpha = (\alpha_{01}, \dots, \alpha_{0k}) \mapsto P_{0j}(\alpha) : \begin{cases} [0, \tau] & \rightarrow [0, 1] \\ t & \mapsto -\frac{(-1 + \exp(\sum_{i=1}^k \alpha_{0i} t)) \alpha_{0j}}{\sum_{i=1}^k \alpha_{0i}} \end{cases} \end{cases}.$$

As this function is differentiable, we obtain the following weak convergence by an application of the functional delta method van der Vaart (1998, p.297).

$$\left\{ \sqrt{n} \left(\hat{P}_{0j}^{(1)}(t) - P_{0j}^{(1)}(t) - (\hat{P}_{0j}^{(2)}(t) - P_{0j}^{(2)}(t)) \right) \right\}_{(t,j) \in \mathcal{M}} \rightsquigarrow \{G_{0j}(t)\}_{(t,j) \in \mathcal{M}} \quad (27)$$

in $\ell^\infty(\mathcal{M})$ where $\{G_{0j}(t)\}_{(t,j) \in \mathcal{M}}$ is a centered Gaussian process. This convergence statement is equivalent to equation (A.7) in Dette et al. Dette et al. (2018) and by applying the functional delta method for directionally Hadamard differentiable maps Carcamo et al. (2020) we get

$$\begin{aligned} \sqrt{n} \left(\max_{(t,j) \in \mathcal{M}} \left| \hat{P}_{0j}^{(1)}(t) - \hat{P}_{0j}^{(2)}(t) \right| - \max_{(t,j) \in \mathcal{M}} \left| P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t) \right| \right) \\ \rightsquigarrow \mathcal{Z} := \max \left\{ \max_{(t,j) \in \mathcal{E}^+} G_{0j}(t), \max_{(t,j) \in \mathcal{E}^-} -G_{0j}(t) \right\}, \end{aligned} \quad (28)$$

where

$$\mathcal{E}^\pm := \left\{ (u, i) \in \mathcal{M} \mid P_{0i}^{(1)}(u) - P_{0i}^{(2)}(u) = \pm \max_{(t,j) \in \mathcal{M}} \left| P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t) \right| \right\}.$$

This statement is analog to the statement of Theorem 3 in Dette et al. Dette et al. (2018).

Recall that $\hat{\alpha}^{*(1)}$ and $\hat{\alpha}^{*(2)}$ are the MLEs of the bootstrap transition intensities defined in Step 3.2 in Algorithm 1 and that $\hat{\hat{\alpha}}^{(1)}$ and $\hat{\hat{\alpha}}^{(2)}$ are the constrained estimators defined in (14). Note that these estimators are MLEs computed by optimizing the log-likelihood function in (6) over the closed parameter space

$$\Theta = \left\{ (\alpha^{(1)}, \alpha^{(2)}) \in \mathbb{R}_{\geq 0}^{2k(k-1)} \mid \max_{j=1}^k \max_{t \in [0, \tau]} \left| P_{0j}^{(1)}(t) - P_{0j}^{(2)}(t) \right| = \varepsilon_\infty \right\}.$$

By Theorem 2 in Wald Wald (1949) it therefore follows that the estimators are consistent, that is

$$\hat{\hat{\alpha}}^{(\ell)} \xrightarrow{\mathbb{P}} \alpha^{(\ell)} \quad (\ell = 1, 2). \quad (29)$$

For the proof of Theorem 1, we further have to show the weak convergence in probability for the bootstrap statistic $\hat{d}_\infty^*$ (for a definition of weak convergence in probability see van der Vaart (1998, p.326ff.)). For this purpose, we note that the bootstrap data is sampled from a different distribution for each sample size n and that it can be shown using the consistency of the constrained estimator in (29) that a conditional version of the Lindeberg-Feller conditions holds Lange (2024, p.96ff.). An application of a corresponding conditional version of the Lindeberg-Feller theorem Lange (2024, p.96ff.) shows that the vector

$$\sqrt{n} \left(\hat{\alpha}^{*(1)} - \hat{\alpha}^{(1)} - \left(\hat{\alpha}^{*(2)} - \hat{\alpha}^{(2)} \right) \right)$$

converges weakly conditionally on the sample $X_1^{(1)}, \dots, X_{n_1}^{(1)}, X_1^{(2)}, \dots, X_{n_2}^{(2)}$ in probability to a multivariate normal distribution with mean 0 and diagonal covariance matrix. Similar to the proof of Theorem 4 in Dette et al. (2018), we can linearize the function

$$t \mapsto \sqrt{n} \left(\hat{P}_{0j}^{*(1)}(t) - \hat{P}_{0j}^{(1)}(t) - \left(\hat{P}_{0j}^{*(2)}(t) - \hat{P}_{0j}^{(2)}(t) \right) \right) \quad (30)$$

by an application of Taylor's theorem. With the consistency of the constrained estimator given in (29), it follows that the function in (30) converges weakly conditionally on the sample in probability to the Gaussian process $\{G_{0j}(t)\}$ defined in (27). This result is analog to equation (A.25) in Dette et al. (2018), and we are now in the position to prove the three parts of Theorem 1.

- (1) To prove (1) we note that the same arguments used to infer (A.32) in Dette et al. (2018) show that $\sqrt{n}(\hat{d}_\infty^* - \hat{d}_\infty)$ is bounded from above by $\mathcal{Z}_n^* + o_{\mathbb{P}}(1)$, where $\mathcal{Z}_n^*$ is a bootstrap statistic converging weakly conditionally on the sample in probability to $\mathcal{Z}$ and $o_{\mathbb{P}}(1)$ is a term converging to 0 in probability for $n \rightarrow \infty$. From there, the proof of (1) is analog to the proof of (4.9) in Theorem 4 in Dette et al. (2018).
- (2) For the proof of (2), we can show that if $\mathcal{E} = \{(t_0, j_0)\}$, then

$$\sqrt{n}(\hat{d}_\infty^* - \hat{d}_\infty) \rightsquigarrow |G_{0j_0}(t_0)|$$

conditionally on the sample in probability. This is equivalent to statement (A.30) in Dette et al. (2018) and the proof follows as the proof of (4.10) in that article.

- (3) The assertion (3) of Theorem 1 follows by the same arguments as given in the derivation of (3.18) in Dette et al. (2018).

□